

Adversarial Security Gate for MLOps Pipelines: A Multi-Dataset Evaluation for IoT/IIoT Intrusion Detection

Anonymous Author 1¹, Anonymous Author 2², Anonymous Author 3¹,
Anonymous Author 4¹, and Anonymous Author 5¹

¹ Anonymous Affiliation 1

² Anonymous Affiliation 2

Abstract. Clean-data validation can promote an intrusion-detection model without testing whether operationally influenceable inputs permit evasion. Prior work studies secure machine-learning lifecycles and realistic adversarial evaluation, but provides limited multi-dataset evidence about connecting both to a frozen, auditable promotion rule. We evaluate such a Security Gate using nine multilayer perceptrons trained with three seeds on independently prepared CICIoT2023, Edge-IIoTset, and ToN_IoT populations. Targeted FGSM and PGD attacks use 1%, 3%, and 5% of train-derived feature ranges, dataset-specific attack masks, bounds, and semantic projection. A validation-only policy combines clean attack recall, worst-case adversarial recall, attack success rate, recall degradation, and integrity checks. All nine candidates pass the clean-only rule, whereas the frozen gate blocks four: all three Edge-IIoTset models and one ToN_IoT model. Mean worst-case attack recall is 0.9864 for CICIoT2023, 0.0008 for Edge-IIoTset, and 0.7793 for ToN_IoT. No semantic violations or split-fingerprint overlaps are recorded. Thus, under the evaluated projected feature-space threat model, adversarial evidence changes promotion eligibility that clean evaluation alone would not question; it does not establish packet-level realizability or guarantee deployment security.

Keywords: SecMLOps · model promotion · constraint-aware evasion · robustness auditing · model governance

1 Introduction

Internet of Things (IoT) and Industrial IoT (IIoT) infrastructures combine heterogeneous devices, protocols, and services, creating a broad and changing attack surface. Learning-based intrusion detection systems (IDSs) can classify patterns that fixed signatures miss, yet a high held-out score is only one condition for operational use. Data preparation, model selection, evaluation, and the decision to release a model must remain reproducible as the system evolves. Machine learning operations (MLOps) organizes this lifecycle through automation, versioning, experiment tracking, and continuous evaluation [11]. Empirical model-

management research also shows that documentation, reproducibility, and versioned assets remain practical concerns after model development [12].

Security adds a distinct requirement. An IDS can satisfy a clean validation threshold while relying on a small set of traffic features that an adversary can influence at inference time. FGSM introduced an efficient first-order adversarial construction [7], and projected-gradient methods iteratively optimize within a bounded region [13]. These attacks provide useful diagnostic tools, but conventional norm bounds were developed largely for perceptual domains. A network-flow record contains integers, categories, immutable identifiers, protocol-dependent fields, and derived quantities. Modifying them independently may create a vector that fools a classifier but does not describe coherent traffic.

Recent evidence makes both the security and realism problems concrete. A study of 139 industrial practitioners found that ML security concerns and defenses are shaped by experienced or anticipated exposure, while also identifying a need for stronger auditing practice [8]. In network intrusion detection, a systematization of knowledge concludes that many adversarial methods do not account for communication-network constraints [21]. Constraint-aware NIDS experiments further show that mutability, value, dependency, and distribution constraints can materially change attack success [1]. Robustness numbers therefore have little operational meaning unless the threat model states what can change, how candidates are projected, and what validity was checked.

Secure MLOps (SecMLOps) addresses the lifecycle side of the problem. It treats security as a cross-cutting concern rather than a post-deployment add-on [23, 24]. Risk-management guidance similarly separates measurement from an organization’s decision about acceptable risk [19, 20]. However, a lifecycle framework does not by itself specify which adversarial measurements should prevent promotion. Conversely, an adversarial benchmark can reveal vulnerability without defining how its evidence enters a controlled release decision. This leaves a practical and scientific question at the intersection of adversarial evaluation, model governance, and MLOps automation.

Our literature comparison identifies a bounded gap: secure ML lifecycles, constrained adversarial examples, IDS defenses, and tabular benchmarks are well studied, but evidence remains limited on applying a frozen, validation-derived promotion policy to constrained results across independent IoT/IIoT datasets. This does not imply that no industrial integration exists; it identifies what the examined peer-reviewed evidence does not evaluate.

This study asks: *How effectively can an automated adversarial Security Gate integrated into an MLOps pipeline identify insufficiently robust intrusion-detection models before production deployment across heterogeneous IoT/IIoT datasets?* Effectiveness is the additional promotion information produced under a fixed threat model and policy. We contribute an auditable pre-promotion gate; a projected, dataset-specific evasion protocol; a three-dataset, three-seed evaluation; and a traceable comparison of clean-only and security-aware decisions and cost. MLOps, FGSM, PGD, and the MLP are established components; their evaluated connection is the contribution.

2 Related Work

Security across the ML lifecycle. MLOps architectures make data, code, model, and deployment transitions explicit [11]. Model-management evidence from 227 repositories highlights maintenance, documentation, versioning, and reproducible environments [12], while security surveys extend concern beyond the classifier to the surrounding software lifecycle [3]. SecMLOps relates people, process, technology, governance, and compliance [23] and maps controls across development and operation [24]. These works locate security responsibilities; our narrower question is what decision changes when an adversarial check controls promotion.

Adversarial robustness in network IDSs. Recent IDS research goes beyond showing that a generic perturbation can reduce accuracy. Adv-Bot generates constrained black-box botnet evasions while preserving malicious functionality in traffic traces [4]. A complementary analysis formalizes mutability, value, dependency, and distribution constraints, then studies attack transferability and defenses [1]. These studies provide stronger problem-space or domain-aware evidence than unrestricted feature perturbation.

Defense work instead changes the detector: AIDTF trains against generated attacks [22], while RAIDS combines reconstruction signals and ensemble classification [16]. We do not repair a candidate after evaluation; we test a frozen candidate against a frozen promotion policy. A gate governs measured risk but does not create robustness.

Constraints in tabular feature space. Tabular robustness requires a threat model that reflects heterogeneous semantics. Cost- and utility-aware formulations show why a uniform distance may not represent an attacker’s capability or incentive [10]. Robust embeddings address categorical features and transfer robustness to model families that do not naturally support gradient-based adversarial training [9]. Constrained Adaptive Attack (CAA) combines gradient and search procedures while enforcing categorical, immutable, and relational constraints [17]. TabularBench subsequently evaluates more than 200 tabular models and several robustification mechanisms with constraint-satisfying attacks [18].

These methods establish that validity constraints affect measured robustness. Our narrower protocol attacks one audited traffic attribute per dataset and projects each step into its encoded domain. It favors an inspectable gate input over attack completeness and cannot bound every feasible network adversary.

Evaluation validity, shift, and governance. The Adversarial Robustness Toolbox supports repeatable attacks [15], but tool consistency does not establish threat-model validity. The NIST taxonomy requires explicit attacker goals, knowledge, capabilities, and lifecycle stage [20]. TableShift separately shows that in-distribution tabular performance does not resolve behavior under distribution

Table 1. Synthesis of related research lines and the residual question addressed here.

Research area	Main evidence	Constraint scope	Residual issue for this study
MLOps/ SecMLOps [11, 24]	Architectures, roles, controls	Full lifecycle; mainly conceptual	No evaluated IDS promotion rule from adversarial metrics
Adversarial NIDS [4, 1]	Realistic evasion and defense experiments	Network and domain constraints	Robustness result not connected to MLOps release state
Tabular robustness [17, 18]	Constraint-aware attacks and benchmarks	Feature validity and relations	Broad benchmarks, not a validation-frozen IDS gate
Model governance	Risk and model-management practices [19, 12]	Documentation and lifecycle control	Risk tolerance remains context-specific
This study	Nine candidates on three IDS datasets	Projected feature constraints and frozen policy	Measures clean-only versus security-aware decision change

shift [6]. We therefore separate within-distribution adversarial evidence from claims about unseen networks.

Governance frameworks recommend measuring, documenting, and managing risk throughout the lifecycle [19]. They deliberately avoid prescribing one universal risk tolerance. Accordingly, our four thresholds instantiate one policy calibrated from validation evidence. PASS means that a candidate is eligible for a later promotion step under that policy; it is neither automatic deployment nor a security guarantee.

Table 1 synthesizes the comparison. The literature provides mature components, but the final column exposes a recurring separation between lifecycle governance and constrained adversarial evidence. Our experiment evaluates this missing connection directly and records the resulting decision delta.

Evaluation datasets. CICIoT2023 captures 33 attacks in a topology of 105 devices [14]. Edge-IIoTset supports centralized and federated IoT/IIoT security experiments [5], and ToN_IoT combines heterogeneous network and telemetry sources [2]. Their collection processes and class structures differ, so agreement across them is informative but does not constitute a pooled population estimate. We prepare, split, train, and interpret every dataset independently.

3 Methodology

3.1 Experimental MLOps Pipeline

Figure 1 separates model construction, policy selection, and final application. Versioned data and feature contracts precede fingerprint-group splitting and

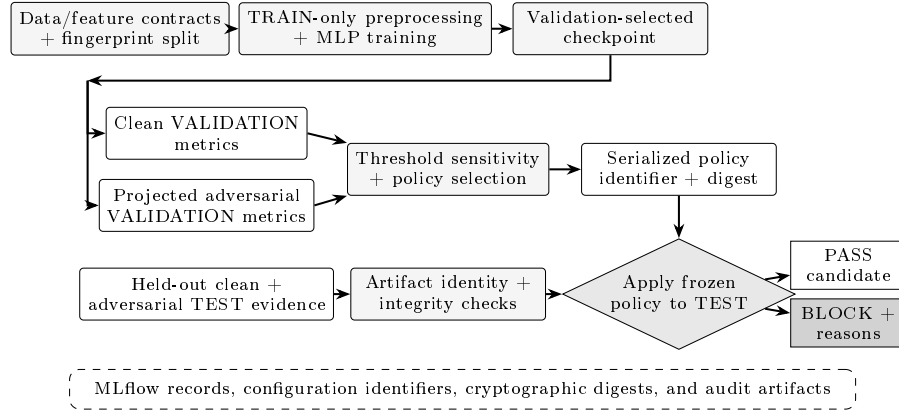


Fig. 1. Evaluated MLOps control flow. Only VALIDATION evidence enters threshold selection; held-out TEST evidence enters when the serialized policy is applied. PASS denotes eligibility for a later promotion process, not deployment.

train-only preprocessing. Clean validation selects a checkpoint; clean and projected-adversarial VALIDATION evidence then informs policy selection. Held-out TEST evidence enters only where the frozen policy is applied, not through the threshold-selection path.

All stochastic model runs use seeds 42, 123, and 2026. No adversarial training is performed. The nine clean models are frozen before adversarial evaluation, and the implemented selector reads only VALIDATION artifacts. The resulting policy is serialized before gate decisions are emitted for TEST artifacts; its identifier and digest remain identical in all nine decision records.

3.2 Datasets and Leakage-Safe Preparation

Table 2 distinguishes source records, candidates remaining after the applicable quality/duplicate policy, and the actual experimental population. CICIOT2023 and Edge-IIoTset target one million observations; Edge reaches 999,996 because complete fingerprint groups are preserved. ToN_IoT uses all 190,474 deduplicated records. A fingerprint is computed from predictive fields before sampling. Cross-label fingerprint groups are excluded, and all copies of a retained fingerprint are assigned to one split. Pairwise TRAIN-VALIDATION-TEST fingerprint overlap is therefore zero. Imputation, scaling, and categorical vocabularies are fitted on TRAIN only and then reused unchanged.

Edge required an explicit data-quality correction. A canonical CSV parser identified four malformed rows among 18,752,710 records. A semantic audit found 158,515 rows with inconsistent numeric fields. The dominant field, `tcp.dstport`, contained 155,549 nonnumeric representations and was removed because no unambiguous semantic normalization was defensible. Residual validation excluded 3,361 rows, leaving 18,749,345

Table 2. Dataset populations, raw feature contracts, and realized 70/15/15 splits.

Dataset	Source N	Candidate N	Exp. N	Raw d	Train	Val.	Test
CICIoT2023	46,776,700	21,055,368 ^a	1,000,000	32	700,000	150,000	150,000
Edge-IIoTset	18,752,710	18,749,345 ^b	999,996	22	699,997	149,999	150,000
ToN_IoT	211,043	190,474 ^c	190,474	20	133,332	28,571	28,571

^aClass-wise deduplicated candidates before excluding cross-label fingerprint groups; ^bcanonical rows after malformed and residual semantic exclusions; ^cunique fingerprints. Raw d is measured after the final feature contract and before categorical expansion.

clean canonical rows and 22 predictive features. The final sample uses group-aware fingerprint assignment and zero overlap. Leakage auditing found no target, metadata, preprocessing, or fingerprint leakage, but strong source/scenario association remained. We therefore retain the exact audit conclusion `NO_DIRECT_LEAKAGE_WITH_DATASET_CONFOUNDING`. Edge results measure discrimination within the sampled capture structure, not cross-scenario generalization.

3.3 Clean IDS Models

Each binary IDS uses the same MLP: 128- and 64-unit hidden layers, rectified linear activations, dropout 0.2, and a two-logit output. Training uses cross-entropy loss, Adam at 10^{-3} , batch size 4096, at most 15 epochs, and early-stopping patience 3. Seeds control initialization and data-loader randomness. The checkpoint with the lowest validation loss is retained. The raw contracts expand to 36, 38, and 50 model inputs for CICIoT2023, Edge-IIoTset, and ToN_IoT, respectively. Test data remain unavailable to fitting and checkpoint selection.

3.4 Projected Constrained Threat Model

The final threat model is `PROJECTED_CONSTRAINED_TABULAR_EVASION`: a white-box adversary knows the model and preprocessing, modifies test-time attack records, and targets `BENIGN` classification. Training data, labels, code, model parameters, categorical indicators, identifiers, and non-masked features are immutable. The initial mutable-continuous-only specification was infeasible because the audited contracts exposed no universally valid continuous feature. We therefore selected one feature per dataset only when it represented an operationally influenceable traffic quantity and admitted deterministic projection. The masks contain `Time_To_Live` for CICIoT2023, `udp.time_delta` for Edge, and `duration` for ToN_IoT. TTL is rounded to its integer domain; timing and duration are clipped to `TRAIN` bounds. These choices deliberately under-approximate attacker capability and are not claimed to span every feasible traffic manipulation.

For mutable feature j and relative budget $b \in \{0.01, 0.03, 0.05\}$,

$$\epsilon_j^{raw} = b(\max_{x \in TRAIN} x_j - \min_{x \in TRAIN} x_j), \quad \epsilon_j^{model} = \epsilon_j^{raw} / s_j, \quad (1)$$

where s_j is the train-fitted standardization scale. FGSM takes one targeted step toward BENIGN. PGD uses 20 iterations, random initialization, step size $\epsilon/4$, and projection after every step. Projection enforces the attack mask, epsilon ball, TRAIN bounds, and the integer domain where applicable. Only ATTACK examples are changed; BENIGN examples remain untouched so specificity remains observable. A condition is valid only when its semantic-violation count is zero. Compliance with these encoded constraints establishes feature-space validity, not the existence of an equivalent packet trace.

A deterministic test subset is selected with seed 314159 and shared across model seeds. CIC and ToN use 20,000 records. Edge preserves selected fingerprint groups, yielding 80,861 records rather than splitting duplicated groups; this larger realized subset is reported rather than treated as 20,000. The 54 test conditions comprise three datasets, three model seeds, two attacks, and three budgets. Every condition completed.

3.5 Security Gate

The gate aggregates evidence in three steps. First, clean and adversarial artifacts must match the candidate checkpoint, feature contract, subset, and configuration. Second, the worst valid value of each robustness criterion is selected independently across the six attack/budget conditions; maximum ASR and minimum adversarial recall need not originate from the same condition. Third, a conjunctive policy emits PASS only when every metric and integrity criterion succeeds. The frozen global thresholds are clean attack recall ≥ 0.95 , worst-case adversarial attack recall ≥ 0.80 , maximum attack success rate (ASR) ≤ 0.20 , and maximum recall drop ≤ 0.20 . Integrity requirements include six completed conditions, zero semantic violations, zero fingerprint overlap, compatible artifacts, and no confirmed leakage. Edge’s confounding status produces a warning but not an automatic block.

Threshold sensitivity uses adversarial VALIDATION artifacts only. The validation-derived policy, `projected-evasion-ids-v1`, has an identifier and cryptographic digest that match all nine TEST decisions. Stable reason codes identify clean, robustness, ASR, degradation, and integrity failures. PASS authorizes only consideration by a later promotion process.

3.6 Metrics and Reproducibility

Clean and adversarial evaluation records accuracy, precision, attack recall, specificity, F1, ROC-AUC, PR-AUC, Matthews correlation coefficient (MCC), false-positive and false-negative rates, and confusion counts. For originally correct attack examples C and adversarial predictions $\hat{y}_i^{adv}$,

$$\text{ASR} = \frac{|\{i \in C : \hat{y}_i^{adv} = \text{BENIGN}\}|}{|C|}, \quad \Delta R = R_{\text{clean}, \text{subset}} - \min_a R_{\text{adv}, a}. \quad (2)$$

The minimum ranges over valid attack/budget conditions. Mean and sample standard deviation use the three model seeds as experimental replicates; rows,

Table 3. Clean full-test performance across three seeds (mean $\pm$ standard deviation).

Dataset	Accuracy	Attack recall	F1	MCC
CICIoT2023	.97820 $\pm$.00001	.98929 $\pm$.00081	.98851 $\pm$.00001	.77568 $\pm$.00307
Edge-IIoTset	1.00000 $\pm$.00000	1.00000 $\pm$.00000	1.00000 $\pm$.00000	1.00000 $\pm$.00000
ToN_IoT	.99085 $\pm$.00093	.99563 $\pm$.00100	.99414 $\pm$.00060	.97333 $\pm$.00269

batches, and attack conditions are not treated as independent replicates. With three seeds, dispersion is descriptive and no significance test is claimed.

Execution used CPU-only PyTorch 2.5.1, scikit-learn 1.5.2, pandas 2.2.3, NumPy 2.1.3, ART 1.18.2, and MLflow 2.18.0 in Docker. Two identifier classes serve different purposes. Predictive-field fingerprints keep duplicate observations within one split and verify zero cross-split overlap. Separately, cryptographic digests bind checkpoints, configurations, epsilon vectors, policies, and result records before their metrics are consumed. Full values remain in structured manifests rather than the article. The study stores metrics and provenance rather than large adversarial arrays, limiting exact sample-level reanalysis.

4 Results

4.1 Clean Performance

Table 3 summarizes full-test clean performance. CIC and ToN show high recall and F1 across seeds. Edge is perfect on its within-distribution test set, but this result remains qualified by dataset confounding. CIC’s lower MCC relative to recall reflects its class distribution and lower specificity; reporting only accuracy or recall would conceal that behavior. These metrics establish that all candidates are plausible under conventional evaluation, which is necessary for assessing whether the gate adds decision information rather than merely rejecting poor clean models.

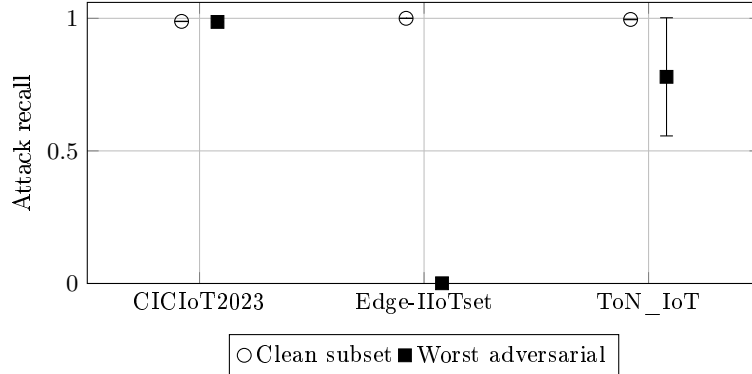
4.2 Adversarial Robustness

Worst-case results in Table 4 answer the central robustness question. CIC remains stable: its mean recall gap is 0.0020 and maximum ASR is approximately 0.0020. Edge changes from perfect clean subset recall to near-zero worst-case recall, with ASR approximately 0.9992. ToN is seed-sensitive; mean worst recall is 0.7793 with standard deviation 0.2228, driven by seed 42 (0.5228) versus seeds 123 (0.8909) and 2026 (0.9242). No condition reports a semantic-constraint violation. Because each column is reduced according to its own worst-case definition, the table does not imply that minimum recall and maximum ASR always arise from the same attack/budget condition.

Figure 2 visualizes this contrast. High clean recall does not imply a small robustness gap: Edge has the strongest clean score and the weakest constrained adversarial score.

Table 4. Worst-case subset robustness across three seeds.

Dataset	Clean recall	Worst adv. recall	Max ASR	Recall gap
CICIoT2023	.9884 $\pm$.0006	.9864 $\pm$.0005	.0020 $\pm$.0001	.0020 $\pm$.0001
Edge-IIoTset	1.0000 $\pm$.0000	.0008 $\pm$.0006	.9992 $\pm$.0006	.9992 $\pm$.0006
ToN_IoT	.9957 $\pm$.0010	.7793 $\pm$.2228	.2174 $\pm$.2239	.2164 $\pm$.2232

**Fig. 2.** Clean subset recall and worst-case adversarial recall across seeds (mean $\pm$ sample standard deviation). Horizontal offsets prevent overlapping markers; ToN's large adversarial dispersion is visible.

4.3 Budget and Attack Comparison

Figure 3 reports mean recall by attack and budget. CIC degradation increases monotonically but remains small. Edge exhibits a threshold-like collapse between 1% and 3% for both attacks. ToN is non-monotonic under FGSM: mean recall increases from 0.8121 at 1% to 0.8761 at 5%. PGD decreases from 0.8199 to 0.7918. Therefore, the data do not support a universal claim that larger budgets monotonically reduce recall.

FGSM and PGD are similar for CIC and Edge at matched budgets. For ToN, neither dominates across all budgets: PGD recall is 0.0078 higher at 1%, then 0.0359 and 0.0842 lower at 3% and 5%. PGD costs substantially more generation time because it uses 20 projected steps. These comparisons are descriptive; three model seeds do not justify strong inferential claims.

4.4 Security Gate Decisions

Table 5 shows the nine decisions. All candidates pass the clean recall floor. The full gate passes all CIC seeds and ToN seeds 123 and 2026, but blocks all Edge seeds and ToN seed 42. Consequently, clean-only evaluation would mark 9/9 candidates as eligible, whereas the full policy marks 5/9 as eligible and blocks 4/9, a decision-change rate of 44.4%. Figure 4 exposes the seed-level pattern: CIC is stable PASS, Edge is stable BLOCK, and ToN is variable. Together,

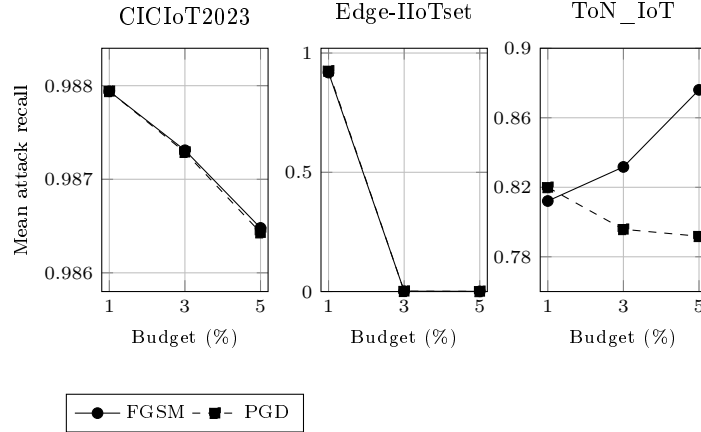


Fig. 3. Attack recall versus train-range-relative budget (three-seed means). Dataset-specific panels avoid compressing CIC and ToN behind Edge’s full-range collapse; vertical scales therefore differ and must not be compared visually without their tick labels.

Table 5. Frozen-policy test decisions. RR, ASR, and Drop are robustness criteria.

Dataset	Seeds	Clean	PASS	BLOCK	Pattern	Failed criteria
CICIoT2023	42/123/2026	3	3	0	stable	none
Edge-IIoTset	42/123/2026	3	0	3	stable	RR, ASR, Drop
ToN_IoT	42/123/2026	3	2	1	seed-varying	RR, ASR, Drop

the table and figure show that the gate identifies four candidates whose clean evidence alone does not reveal policy-defined insufficient robustness.

Edge’s blocks arise from measured robustness criteria, not from its confounding warning: each seed fails robust recall, ASR, and recall-drop thresholds. The warning remains necessary because perfect clean performance may exploit scenario/source associations. ToN’s mixed outcome reveals sensitivity to model initialization that its narrow clean standard deviation did not expose.

4.5 Operational Overhead

Table 6 separates model training from the sum of attack-generation runtimes across six conditions per seed. Edge’s raw suite time is higher because its group-preserving subset contains 80,861 records versus 20,000 for CIC and ToN. Normalizing by sample-condition produces 12.43–13.51 ms per 1,000 records, so the artifacts do not support an efficiency ranking among datasets. The values exclude data preparation, artifact validation, orchestration, and production serving. Adversarial evaluation and gate logic were not instrumented separately at sufficient precision and are not assigned invented costs.

	Model seed			
	42	123	2026	
CICIoT2023	P	P	P	P = PASS B = BLOCK
Edge-IIoTset	B	B	B	
ToN_IoT	B	P	P	

Fig. 4. Frozen-policy Security Gate decisions by dataset and model seed. Every candidate passed clean-only recall; the symbols show eligibility under this evaluated policy, not general dataset security.

Table 6. Measured CPU time per seed (mean $\pm$ standard deviation) and workload-normalized attack generation. Each attack suite contains six conditions.

Dataset	Subset N	Training (s)	Suite (s)	ms/1k/cond.	Suite/train
CICIoT2023	20,000	256.66 $\pm$ 37.74	1.55 $\pm$ 0.04	12.92	0.6%
Edge-IIoTset	80,861	225.85 $\pm$ 66.14	6.03 $\pm$ 1.04	12.43	2.7%
ToN_IoT	20,000	46.47 $\pm$ 7.87	1.62 $\pm$ 0.19	13.51	3.5%

5 Discussion

Decision value beyond clean evaluation. The principal finding is that a deterministic pre-promotion control changes decisions that clean evaluation accepts: four of nine candidates move from clean PASS to security BLOCK. CIC is the necessary counterexample—the gate does not reject every attacked model, because all three CIC candidates retain about 0.986 worst-case recall. Edge shows the opposite outcome, and ToN shows seed dependence despite small clean variation. The policy therefore uses robustness evidence rather than proxying clean performance.

This operationalizes SecMLOps traceability [23, 24]: checkpoint, adversarial bundle, policy digest, and reason codes must agree before eligibility changes. It makes one technical decision reviewable without replacing organizational risk management, consistent with the AI RMF distinction between measurement and context-specific tolerance [19].

Dataset-specific evidence. For CICIoT2023, every seed passes under the TTL-only attack and the mean recall gap is about 0.002. This shows only that the permitted TTL change does not cross the policy; it says nothing about other features, black-box attacks, poisoning, or shift.

For Edge-IIoTset, perfect clean discrimination coexists with almost complete adversarial collapse, and every seed violates robust-recall, ASR, and degradation limits. Sensitivity along `udp.time_delta` is a hypothesis, not a causal result. The audit conclusion `NO_DIRECT_LEAKAGE_WITH_DATASET_CONFOUNDING` says that audited direct channels were absent, not that confounding disappeared. Robustness determines BLOCK; confounding separately limits external interpretation.

For ToN_IoT, seed 42 is blocked while 123 and 2026 pass. Clean attack recall varies by about one tenth of a percentage point, but worst-case recall spans 0.5228–0.9242. The gate exposes candidate-level instability hidden by clean performance; because optimization paths were not isolated, initialization is not claimed as the sole cause.

Constraints, attacks, and budget response. Zero semantic violations support correct execution of the encoded rules. This agrees with NIDS and tabular studies showing that constraints affect attack effectiveness [1, 18]. Our evidence is weaker than Adv-Bot’s problem-space reasoning about malicious functionality [4]: stored feature vectors do not prove realizable traffic.

FGSM and PGD have similar matched-budget effects for CIC and Edge; for ToN, PGD is stronger at 3% and 5% but weaker at 1%. CIC degrades monotonically, Edge collapses between 1% and 3%, and ToN FGSM improves with budget. Projection, clipping, scaling, or decision geometry may contribute, but were not isolated. Thus, the gate uses the worst valid observation instead of assuming that the largest budget or iterative attack is strongest. Adaptive constrained methods such as CAA [17] belong in a future pre-registered protocol.

Defense work offers remediation but answers another question: AIDTF uses adversarial training [22], RAIDS combines reconstruction and ensembles [16], and robust embeddings transfer protection across tabular models [9]. The gate selects none of them; it records noncompliance for a later engineering decision.

Operational interpretation. Attack-suite time is 0.6–3.5% of mean training time, and normalized values are similar despite Edge’s larger subset. Attack generation was not dominant in this CPU experiment, but data access, orchestration, failure recovery, and serving were not timed. Production overhead and validation frequency remain deployment-specific.

Clean and adversarial evaluation answer different questions. A frozen gate makes that difference visible when lifecycle state can change. PASS remains conditional on model, data, mask, budgets, and thresholds; BLOCK means policy noncompliance, not universal failure.

6 Threats to Validity

Internal validity. Deterministic data fingerprints, group-aware splits, train-only preprocessing, frozen checkpoints, and cryptographic artifact digests reduce contamination and post-selection risk. Cross-label fingerprint groups are excluded before splitting. Edge required a parser and semantic correction; the canonical count, four malformed rows, removal of `tcp.dstport`, and residual exclusions remain in the audit trail. Nevertheless, source labels, capture procedures, or feature extraction may contain errors not detectable by these checks. The large amount of duplication in CIC and scenario association in Edge also mean that zero split overlap is necessary but not sufficient evidence of independent real-world behavior.

Construct validity. The experiment measures targeted projected feature-space evasion, not verified packet-level attacks on live networks. The one-feature masks are conservative and reproducible, but likely under-approximate some attackers. Conversely, changing a tabular duration or timing value within bounds does not prove that an equivalent packet trace is realizable while preserving malicious functionality. Train-range-relative budgets depend on observed extrema and do not express equal operational effort across datasets. Semantic validity therefore means compliance with encoded constraints, not complete protocol realization or attacker cost.

External validity. Three public datasets, one MLP architecture, two white-box attacks, and three budgets do not cover all networks, models, or adversaries. Each dataset is evaluated independently; the study does not estimate a common deployment population. Edge’s scenario confounding particularly limits transfer claims, and TableShift’s evidence on tabular distribution shift reinforces why in-distribution results should not be generalized automatically [6]. Temporal and cross-site data, tree and transformer models, black-box or adaptive attacks, and problem-space replay remain unevaluated.

Conclusion validity. The three model seeds are the only experimental replicates. Means, sample standard deviations, and decision counts are descriptive; samples and attack conditions are not promoted to pseudo-replicates, and no significance claim is made. The 44.4% decision-change rate characterizes these nine candidates only. Thresholds encode a validation-derived policy, not a universal optimum or risk tolerance. The selector reads validation artifacts and the decision records share its digest, but the repository does not prove prospective freezing before TEST attacks were generated. Sensitivity analysis reduces arbitrary selection but cannot remove policy dependence. ToN’s seed-sensitive result makes replication particularly important.

7 Conclusions and Future Work

An automated adversarial Security Gate identified policy-defined insufficient robustness that clean evaluation did not expose. Under the frozen projected constrained threat model, all nine candidates passed clean recall, but four (44.4%) were blocked after worst-case robustness, ASR, degradation, and integrity checks. CIC remained within every threshold across seeds, Edge was consistently blocked despite perfect clean metrics, and ToN was seed-sensitive. The research question therefore has a conditional affirmative answer: across these three datasets and nine MLP candidates, a reproducible gate adds actionable promotion evidence that clean validation alone omits.

The scope of that answer is defined by a white-box feature-space adversary, one audited feature per dataset, three budgets, two attacks, and validation-derived thresholds. The gate does not guarantee security, and PASS is eligibility rather than deployment. Future work should validate packet-level realizability,

expand model families and adaptive constrained attacks, evaluate temporal and cross-network shifts, and prospectively register policy calibration before final test generation and evaluation.

References

1. Alhussien, N., Aleroud, A., Melhem, A., Khamaiseh, S.Y.: Constraining adversarial attacks on network intrusion detection systems: Transferability and defense analysis. *IEEE Transactions on Network and Service Management* **21**(3), 2751–2772 (2024). <https://doi.org/10.1109/TNSM.2024.3357316>
2. Alsaedi, A., Moustafa, N., Tari, Z., Mahmood, A.N., Anwar, A.: TON_IoT telemetry dataset: A new generation dataset of IoT and IIoT for data-driven intrusion detection systems. *IEEE Access* **8**, 165130–165150 (2020). <https://doi.org/10.1109/ACCESS.2020.3022862>
3. Chen, H., Babar, M.A.: Security for machine learning-based software systems: A survey of threats, practices, and challenges. *ACM Computing Surveys* **56**(6), 1–38 (2024). <https://doi.org/10.1145/3638531>
4. Debicha, I., Cochez, B., Kenaza, T., Debatty, T., Dricot, J.M., Mees, W.: Adv-Bot: Realistic adversarial botnet attacks against network intrusion detection systems. *Computers & Security* **129**, 103176 (2023). <https://doi.org/10.1016/j.cose.2023.103176>
5. Ferrag, M.A., Friha, O., Hamouda, D., Maglaras, L., Janicke, H.: Edge-IIoTset: A new comprehensive realistic cyber security dataset of IoT and IIoT applications for centralized and federated learning. *IEEE Access* **10**, 40281–40306 (2022). <https://doi.org/10.1109/ACCESS.2022.3165809>
6. Gardner, J., Popovic, Z., Schmidt, L.: Benchmarking distribution shift in tabular data with TableShift. In: *Advances in Neural Information Processing Systems* 36 (2023). <https://doi.org/10.52202/075280-2324>
7. Goodfellow, I.J., Shlens, J., Szegedy, C.: Explaining and harnessing adversarial examples. In: *International Conference on Learning Representations* (2015), <https://arxiv.org/abs/1412.6572>
8. Grosse, K., Bieringer, L., Besold, T.R., Biggio, B., Krombholz, K.: Machine learning security in industry: A quantitative survey. *IEEE Transactions on Information Forensics and Security* **18**, 1749–1762 (2023). <https://doi.org/10.1109/TIFS.2023.3251842>
9. Kireev, K., Andriushchenko, M., Troncoso, C., Flammarion, N.: Transferable adversarial robustness for categorical data via universal robust embeddings. In: *Advances in Neural Information Processing Systems* 36 (2023). <https://doi.org/10.52202/075280-0472>
10. Kireev, K., Kulynych, B., Troncoso, C.: Adversarial robustness for tabular data through cost and utility awareness. In: *Network and Distributed System Security Symposium* (2023). <https://doi.org/10.14722/ndss.2023.24924>
11. Kreuzberger, D., Kuehl, N., Hirschl, S.: Machine learning operations (MLOps): Overview, definition, and architecture. *IEEE Access* **11**, 31866–31879 (2023). <https://doi.org/10.1109/ACCESS.2023.3262138>
12. Latendresse, J., Abedu, S., Abdellatif, A., Shihab, E.: An exploratory study on machine learning model management. *ACM Transactions on Software Engineering and Methodology* **34**(1), 1–31 (2025). <https://doi.org/10.1145/3688841>

13. Madry, A., Makelov, A., Schmidt, L., Tsipras, D., Vladu, A.: Towards deep learning models resistant to adversarial attacks. In: International Conference on Learning Representations (2018), <https://openreview.net/forum?id=rJzIBfZAb>
14. Neto, E.C.P., Dadkhah, S., Ferreira, R., Zohourian, A., Lu, R., Ghorbani, A.A.: CICIOT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment. *Sensors* **23**(13), 5941 (2023). <https://doi.org/10.3390/s23135941>
15. Nicolae, M.I., et al.: Adversarial robustness toolbox v1.0.0. CoRR **abs/1807.01069** (2019), <https://arxiv.org/abs/1807.01069>
16. Sarikaya, A., Kilic, B.G., Demirci, M.: RAIDS: Robust autoencoder-based intrusion detection system model against adversarial attacks. *Computers & Security* **135**, 103483 (2023). <https://doi.org/10.1016/j.cose.2023.103483>
17. Simonetto, T., Ghamizi, S., Cordy, M.: Constrained adaptive attack: Effective adversarial attack against deep neural networks for tabular data. In: Advances in Neural Information Processing Systems 37 (2024). <https://doi.org/10.52202/079017-0873>
18. Simonetto, T., Ghamizi, S., Cordy, M.: TabularBench: Benchmarking adversarial robustness for tabular deep learning in real-world use-cases. In: Advances in Neural Information Processing Systems 37 (2024). <https://doi.org/10.52202/079017-2492>
19. Tabassi, E.: Artificial intelligence risk management framework (AI RMF 1.0). Tech. Rep. NIST AI 100-1, National Institute of Standards and Technology (2023). <https://doi.org/10.6028/NIST.AI.100-1>
20. Vassilev, A., Oprea, A., Fordyce, A., Anderson, H., Davies, X., Hamin, M.: Adversarial machine learning: A taxonomy and terminology of attacks and mitigations. Tech. Rep. NIST AI 100-2e2025, National Institute of Standards and Technology (2025). <https://doi.org/10.6028/NIST.AI.100-2e2025>
21. Vitorino, J., Praca, I., Maia, E.: Sok: Realistic adversarial attacks and defenses for intelligent network intrusion detection. *Computers & Security* **134**, 103433 (2023). <https://doi.org/10.1016/j.cose.2023.103433>
22. Xiong, W.D., Luo, K.L., Li, R.: AIDTF: Adversarial training framework for network intrusion detection. *Computers & Security* **128**, 103141 (2023). <https://doi.org/10.1016/j.cose.2023.103141>
23. Zhang, X., Jaskolka, J.: Conceptualizing the secure machine learning operations (SecMLOps) paradigm. In: 2022 IEEE 22nd International Conference on Software Quality, Reliability and Security. pp. 127–138 (2022). <https://doi.org/10.1109/QRS57517.2022.00023>
24. Zhang, X., Zhao, P., Jaskolka, J., Li, H., Lu, R.: SecMLOps: A comprehensive framework for integrating security throughout the machine learning operations lifecycle. *Empirical Software Engineering* **31**(74) (2026). <https://doi.org/10.1007/s10664-025-10795-y>